

Global Fitness Convergence?

A Cross-Country Analysis of Fitness Content in Trending Videos

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Abstract

Social-media platforms circulate workout routines, body ideals, nutrition advice, and wellness practices across national borders. This project asks whether fitness-related trending content shows similar thematic structure across countries. Video titles are grouped into four interpretable dimensions and each country is represented by a topic-prevalence vector. Cosine similarity measures the alignment between country profiles, while bootstrap resampling evaluates the stability of those comparisons. An additional logistic-regression experiment examines whether simple title-level features contain predictive information about relatively high engagement. In the simulation analysis, the three country profiles show substantial pairwise similarity, with the strongest point similarity between India and Pakistan. The bootstrap results indicate considerable sampling uncertainty, and the engagement model provides only moderate predictive performance. Together, the results show how a qualitative question about cultural similarity can be translated into a compact quantitative framework while retaining explicit measures of uncertainty.

1 Introduction

Fitness culture is increasingly shaped by content that travels across national boundaries. Strength training, body transformation, high-protein diets, supplementation, yoga, and mobility practices can all become widely visible through trending-video systems. This raises a natural quantitative question: *to what extent do different countries exhibit similar fitness-content profiles?*

The main difficulty is that “fitness culture” is not directly observed as a single variable. It must first be represented through measurable features. This study therefore focuses on the thematic composition of fitness-related trending videos. Titles are classified into four broad categories—workout, body, nutrition, and wellness—and their prevalence is used to construct a country-level representation. Similarity between countries can then be measured mathematically rather than described only through examples.

The analysis addresses three linked questions. First, how prominent is fitness-related content within each country’s sample? Second, how does the thematic composition of that content differ across countries? Third, how stable are the resulting similarity estimates when the available videos are resampled? A small predictive experiment is included to examine whether country and title-level features provide useful information about video engagement.

2 Data and Measurement

The study uses video-level records containing title, country, views, likes, and comments. The simulation sample contains 36 observations across Germany (DE), India (IN), and Pakistan (PK), with 12 videos per country. After removing invalid or duplicated observations, fitness-related content is identified directly from title text.

Four topic groups are used. **Workout** covers terms related to exercise, gym training, strength, cardio, and running. **Body** captures weight loss, fat loss, muscle, bodybuilding, and body transformation. **Nutrition** includes protein, supplements, diet, calories, and meal preparation. **Wellness** includes yoga, Pilates, stretching, meditation, and mobility. A title may activate more

than one topic, which allows mixed content such as a high-protein workout video to contribute to multiple dimensions.

For descriptive analysis, engagement is defined as

$$\text{EngagementRate} = \frac{\text{Likes} + \text{Comments}}{\text{Views}}.$$

This measure is used only as a relative indicator within the sample.

3 Mathematical Model

3.1 Country topic profiles

For country c and topic k , define

$$p_{c,k} = \frac{\text{number of fitness videos in country } c \text{ assigned to topic } k}{\text{number of fitness-related videos in country } c}.$$

The corresponding country profile is

$$\mathbf{p}_c = (p_{c,\text{workout}}, p_{c,\text{body}}, p_{c,\text{nutrition}}, p_{c,\text{wellness}}).$$

Because one video can belong to several topics, the components need not sum to one. The vector records thematic prevalence rather than a probability distribution over mutually exclusive categories.

3.2 Cross-country similarity

For countries a and b , cosine similarity is

$$S(a, b) = \frac{\mathbf{p}_a^\top \mathbf{p}_b}{\|\mathbf{p}_a\|_2 \|\mathbf{p}_b\|_2}.$$

A value near one indicates that the two countries emphasize the four topics in similar proportions. This measure is appropriate here because the objective is to compare profile composition rather than raw video volume.

3.3 Stability by bootstrap resampling

To assess sampling sensitivity, fitness-related videos are resampled with replacement within each country. A new topic vector and a new cosine similarity are computed in each resample. Repeating this procedure 300 times gives an empirical distribution for each country pair. The 2.5th and 97.5th percentiles form a 95% bootstrap interval.

3.4 Exploratory engagement model

A video is labeled as high engagement when its engagement rate exceeds the median within its country. Logistic regression uses country, fitness-related status, title length, and title word count as predictors. Performance is evaluated with accuracy, balanced accuracy, ROC-AUC, and a confusion matrix. This experiment is descriptive and predictive rather than causal.

4 Simulation Analysis

4.1 Fitness prevalence and topic composition

Table 1 summarizes the sample. Each country contains 12 videos, of which nine are classified as fitness-related, giving a 75% fitness share in all three cases. Mean views are highest for IN, while mean engagement is highest for PK.

Table 1: Country-level summary generated by the pipeline. Values are from the synthetic smoke-test fixture.

Country	Videos	Fitness videos	Fitness share	Mean views	Mean engagement
DE	12	9	75.0%	147,500	4.76%
IN	12	9	75.0%	309,583	5.31%
PK	12	9	75.0%	173,083	5.97%

Figure 1 shows that the overall fitness share alone does not separate the three country samples. The more informative comparison therefore comes from the internal composition of the fitness-related videos.

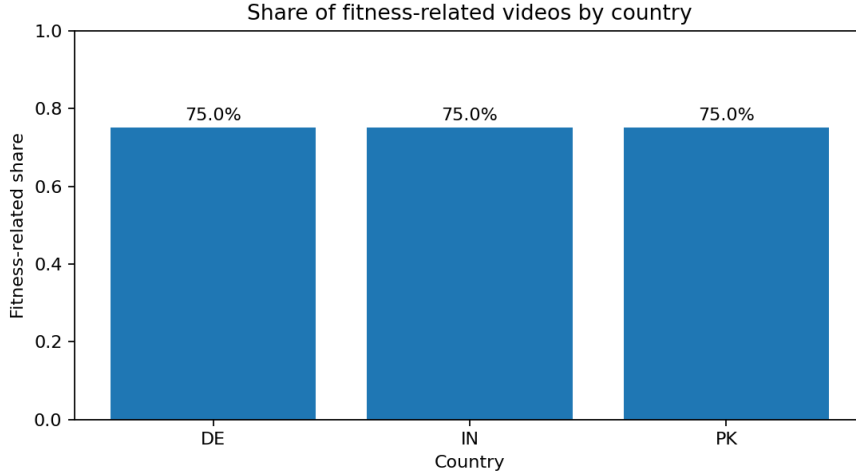


Figure 1: Share of videos classified as fitness-related.

Table 2 and Figure 2 present the four-dimensional topic profiles. DE assigns relatively more weight to nutrition and wellness, both at 0.333, and less to workout content, at 0.222. IN is more concentrated in workout content, with a prevalence of 0.556. PK also emphasizes workout content, at 0.444. The body dimension is identical for IN and PK at 0.222, while DE is lower at 0.111.

Table 2: Country-level fitness topic profiles. Each entry is the share of fitness-related videos assigned to the corresponding topic.

Country	Workout	Body	Nutrition	Wellness
DE	0.222	0.111	0.333	0.333
IN	0.556	0.222	0.222	0.222
PK	0.444	0.222	0.222	0.222

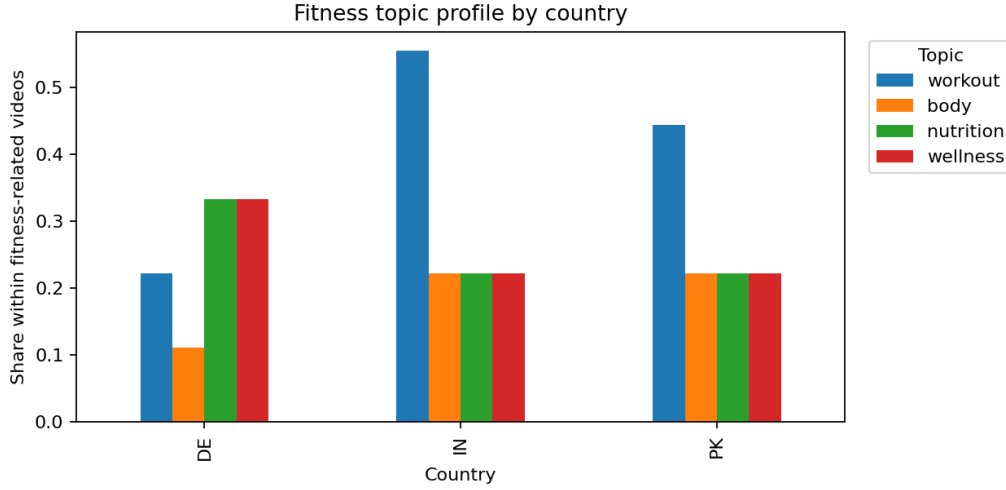


Figure 2: Four-dimensional topic profiles for fitness-related videos.

The comparison illustrates why a multidimensional representation is useful. Countries can have the same overall prevalence of fitness content while differing in what that content emphasizes.

4.2 Cross-country similarity

The cosine-similarity matrix in Table 3 shows that all three pairs have positive and relatively high profile similarity. IN and PK have the strongest point similarity at 0.994. DE–PK follows at 0.867, and DE–IN at 0.823.

Table 3: Cosine similarity between country topic profiles.

Country	DE	IN	PK
DE	1.000	0.823	0.867
IN	0.823	1.000	0.994
PK	0.867	0.994	1.000

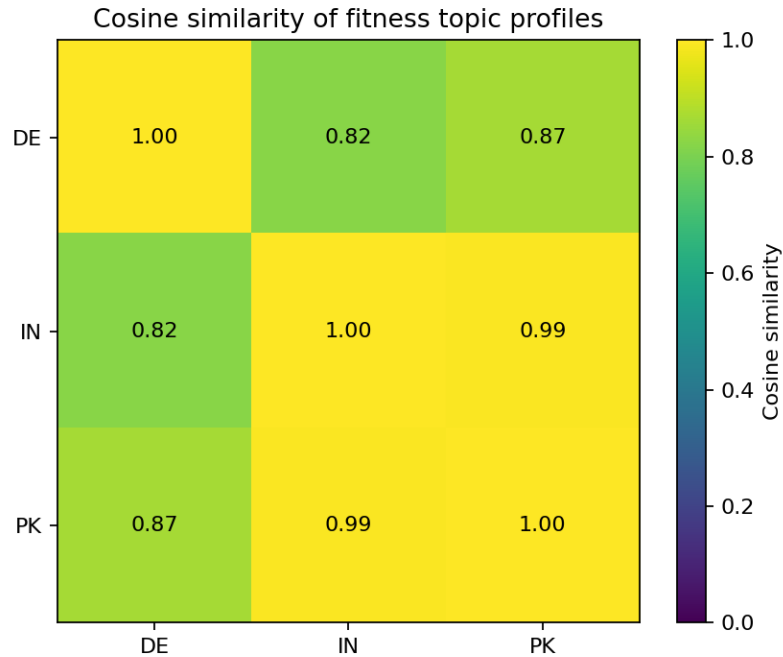


Figure 3: Cosine-similarity matrix for the country topic profiles.

The heatmap in Figure 3 reflects the same structure. IN and PK point in almost the same direction in the four-dimensional topic space, whereas DE differs mainly through its greater relative emphasis on nutrition and wellness.

4.3 Stability of similarity estimates

Point similarity can look stronger than the available sample justifies. Table 4 reports the bootstrap results. The mean resampled similarities are 0.686 for DE–IN, 0.714 for DE–PK, and 0.828 for IN–PK. The corresponding intervals are wide: DE–IN spans 0.263–0.979, DE–PK spans 0.269–0.981, and IN–PK spans 0.465–0.979.

Table 4: Bootstrap similarity estimates based on 300 resamples.

Pair	n_a	n_b	Mean similarity	95% interval
DE–IN	9	9	0.686	[0.263, 0.979]
DE–PK	9	9	0.714	[0.269, 0.981]
IN–PK	9	9	0.828	[0.465, 0.979]

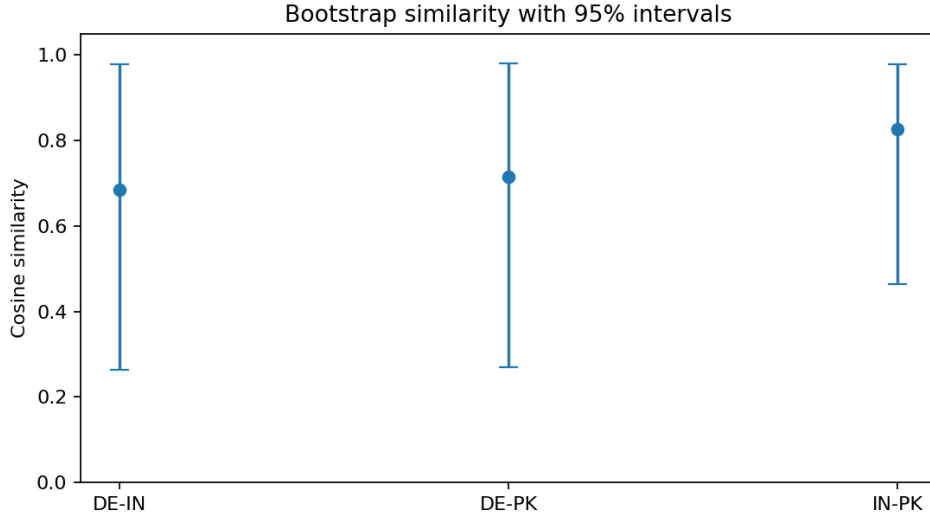


Figure 4: Bootstrap mean similarities with empirical 95% intervals.

Figure 4 makes the uncertainty visible. The ordering of the mean similarities is consistent with the original matrix, with IN–PK remaining the closest pair, but the intervals overlap substantially. The result shows that similarity should be interpreted together with its stability: the profile direction can appear highly aligned even when a small change in the sampled videos noticeably changes the estimate.

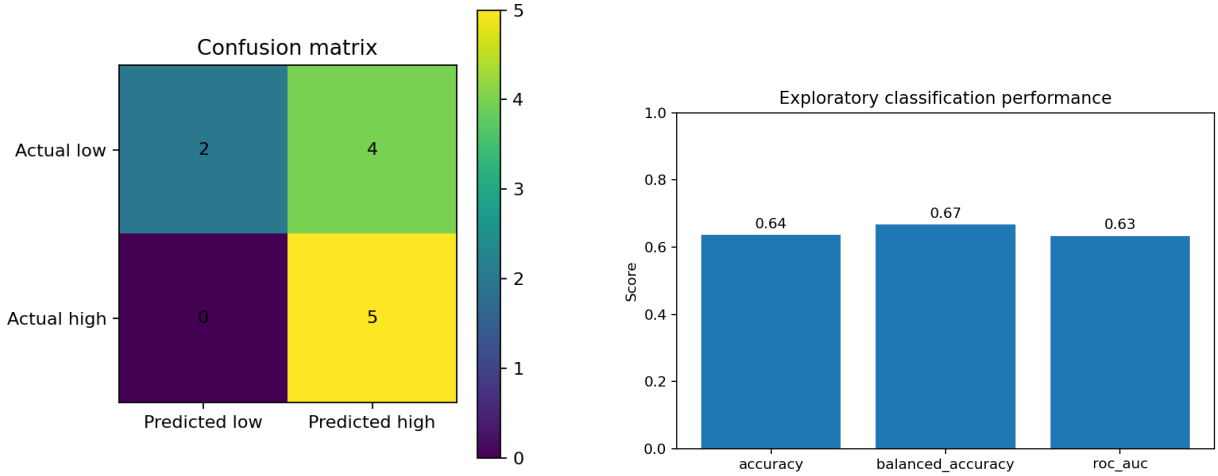
4.4 Engagement prediction

The logistic-regression experiment uses 25 observations for training and 11 for testing. Table 5 reports an accuracy of 0.636, balanced accuracy of 0.667, and ROC–AUC of 0.633.

Table 5: Exploratory logistic-regression performance on the smoke-test fixture.

Metric	Value
Accuracy	0.636
Balanced accuracy	0.667
ROC-AUC	0.633
Training observations	25
Test observations	11
TN / FP / FN / TP	2 / 4 / 0 / 5

The confusion matrix contains two true negatives, four false positives, zero false negatives, and five true positives. The classifier identifies every high-engagement case in the test set, but it also incorrectly assigns high engagement to four of the six low-engagement cases.



(a) Confusion matrix for the exploratory high-engagement classifier.

(b) Accuracy, balanced accuracy, and ROC-AUC for the exploratory classifier.

Figure 5: Evaluation results for the exploratory high-engagement classification model.

The three scores in Figure 5b are only moderately above chance-level discrimination. In this sample, the simple predictors therefore contain some signal but do not support a strong engagement model. This result is useful because it separates the descriptive question of cross-country similarity from the predictive question of engagement: a clear thematic structure does not automatically imply strong predictability at the individual-video level.

5 Conclusion

This study translates a broad question about global fitness culture into a compact mathematical comparison of trending-video content. Country-level topic vectors reveal differences that are hidden by a single fitness-share statistic, while cosine similarity provides an interpretable measure of profile alignment. The simulation results show the strongest thematic similarity between IN and PK, but bootstrap resampling also demonstrates that similarity estimates can be unstable when only a small number of fitness-related videos are available. The engagement experiment examines whether simple observable features can help predict which videos receive relatively high engagement. The model achieves only moderate performance, indicating that country and basic title characteristics alone are not sufficient to explain most of the differences in video engagement.